

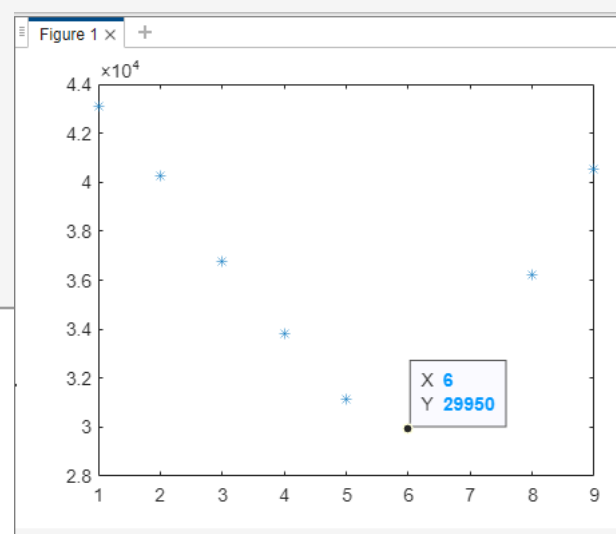
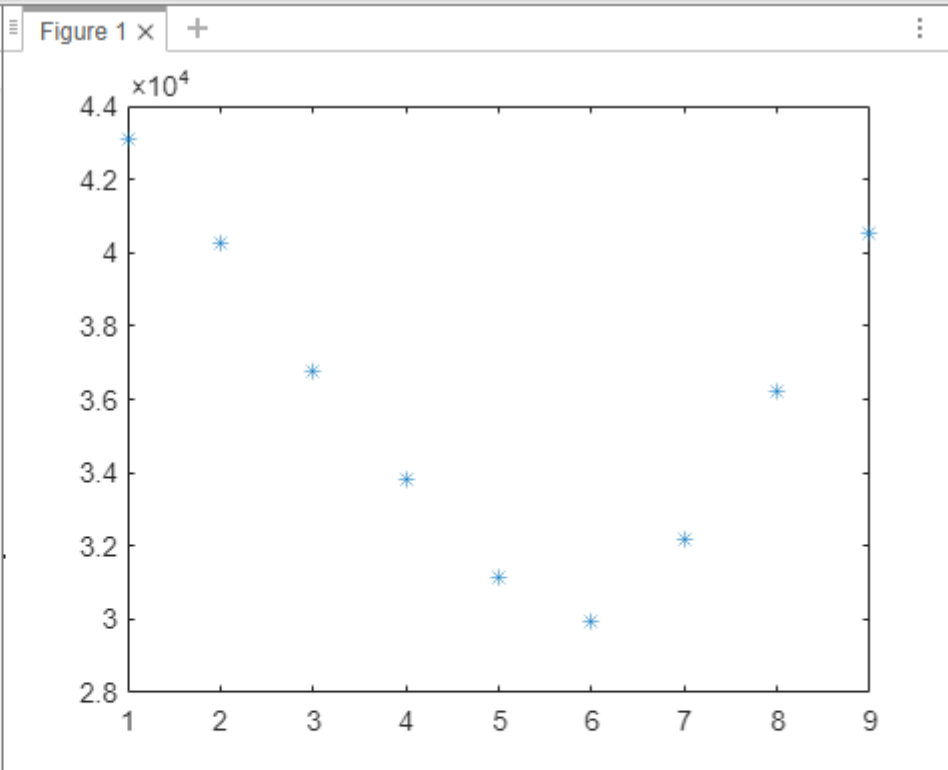
Project 5.5.PJ4

Here, we will find the optimum values for teacher pay rates and number pools in Los Angeles School District.

- In the first scenario, pay rates will be set to substitute teachers earning \$45 and full-time teachers earning \$81.
- The range of the number of teachers from (0,1000).
- The focus will be on Tuesdays.
- We'll find the least expected costs.
- Assumptions: All teachers work the same hours throughout, and there's no overtime. You can't hire all substitute teachers to replace full-time teachers.
- We'll increase the number of simulations to fine-tune the model.

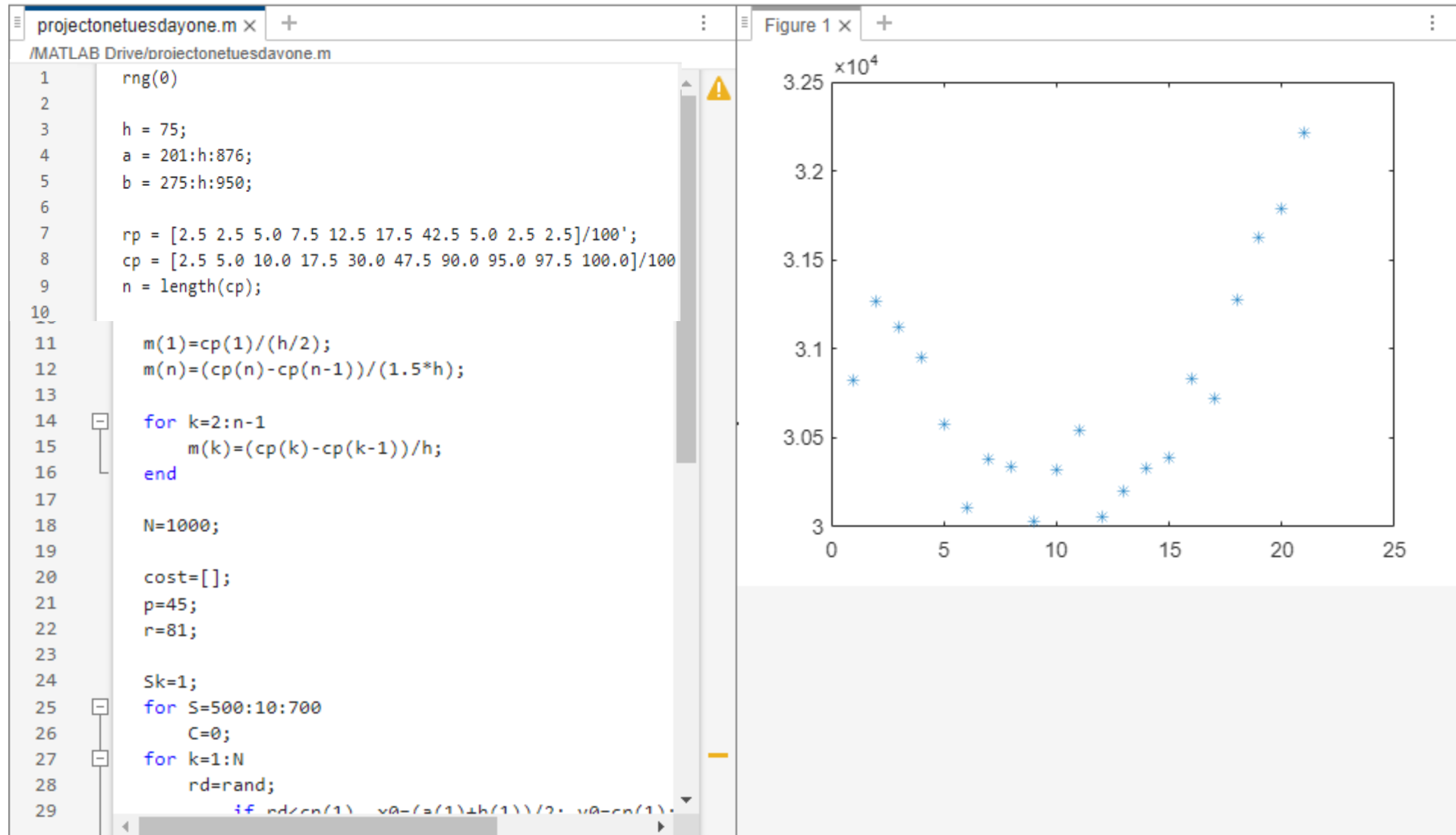
```
projectonetuesdayone.m x +
/MATLAB Drive/projectonetuesdayone.m

1  rng(0)
2
3  h = 75;
4  a = 201:h:876;
5  b = 275:h:950;
6
7  rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8  cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9  n = length(cp);
10
11  m(1)=cp(1)/(h/2);
12  m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14  for k=2:n-1
15      m(k)=(cp(k)-cp(k-1))/h;
16  end
17
18  N=1000;
19
20  cost=[];
21  p=45;
22  r=81;
23
24  Sk=1;
25  for S=100:100:900
26      C=0;
27      for k=1:N
28          rd=rand;
29          if rd/cp(1) <= v0-(=(1)+h(1))/2: v0-cp(1).
```



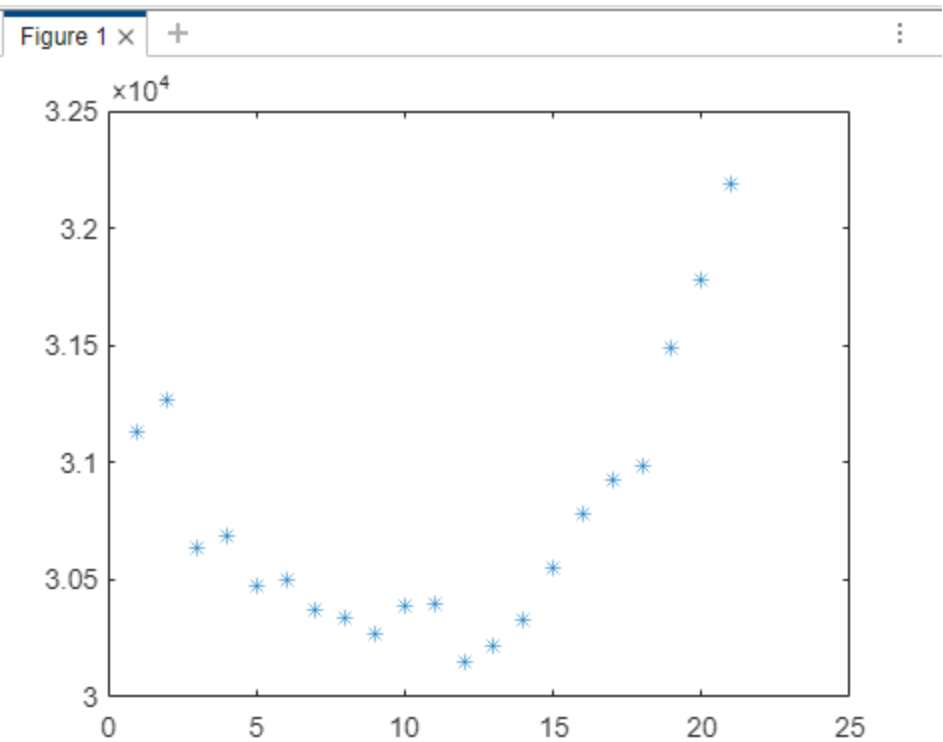
Number of
simulations: 1000

Changing the interval to 200



```
projectonuesdayone.m x +
/MATI AR Drive/nriectonuesdayone m
1 rng(0)
2
3 h = 75;
4 a = 201:h:876;
5 b = 275:h:950;
6
7 rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8 cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9 n = length(cp);
10
11 m(1)=cp(1)/(h/2);
12 m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14 for k=2:n-1
15     m(k)=(cp(k)-cp(k-1))/h;
16 end
17
18 N=2000;
19
20 cost=[];
21 p=45;
22 r=81;
23
24 Sk=1;
25 for S=500:10:700
26     C=0;
27     for k=1:N
28         rd=rand;
29         if rd<cp(1), x0=(a(1)+b(1))/2; y0=cp(1);
30         elseif rd<cp(2), x0=(a(2)+b(2))/2; y0=cp(2);
31         elseif rd<cp(3), x0=(a(3)+b(3))/2; y0=cp(3);
32         elseif rd<cp(4), x0=(a(4)+b(4))/2; y0=cp(4);
33         elseif rd<cp(5), x0=(a(5)+b(5))/2; y0=cp(5);
34         elseif rd<cp(6), x0=(a(6)+b(6))/2; y0=cp(6);
35         elseif rd<cp(7), x0=(a(7)+b(7))/2; y0=cp(7);
36         elseif rd<cp(8), x0=(a(8)+b(8))/2; y0=cp(8);
37         elseif rd<cp(9), x0=(a(9)+b(9))/2; y0=cp(9);
```

Number of
simulations: 2000

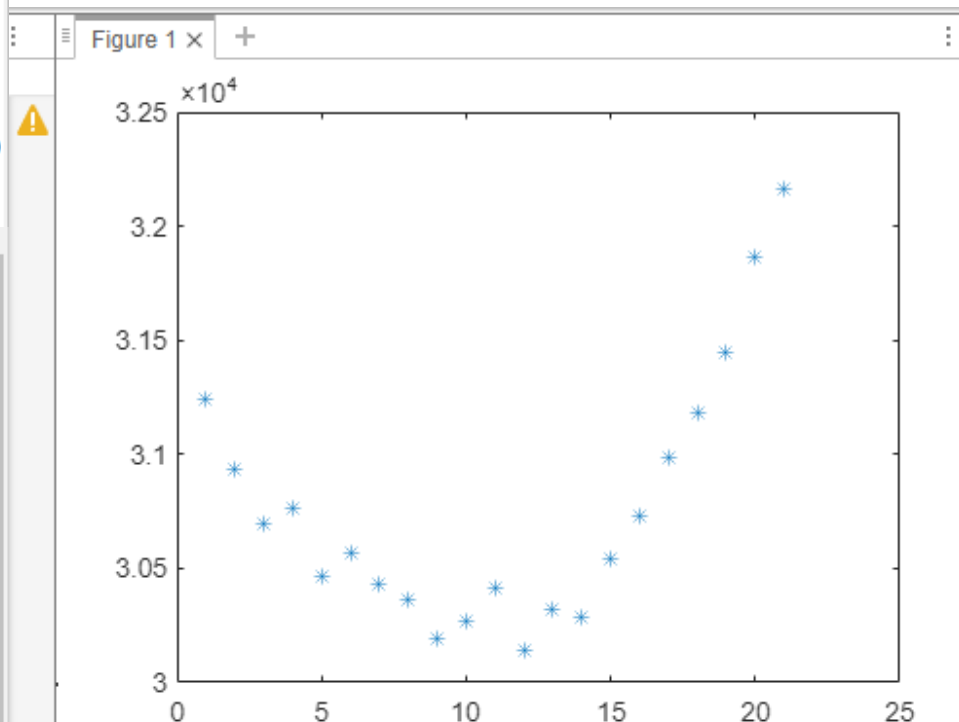


```

1  rng(0)
2
3  h = 75;
4  a = 201:h:876;
5  b = 275:h:950;
6
7  rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8  cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9  n = length(cp);
10
11  m(1)=cp(1)/(h/2);
12  m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14  for k=2:n-1
15      m(k)=(cp(k)-cp(k-1))/h;
16  end
17
18  N=3000;
19
20  cost=[];
21  p=45;
22  r=81;
23
24  Sk=1;
25  for S=500:10:700
26      C=0;
27      for k=1:N
28          rd=rand;
29          if rd<cp(1), x0=(a(1)+b(1))/2; y0=cp(1);
30          elseif rd<cp(2), x0=(a(2)+b(2))/2; y0=cp(2);
31          elseif rd<cp(3), x0=(a(3)+b(3))/2; y0=cp(3);
32          elseif rd<cp(4), x0=(a(4)+b(4))/2; y0=cp(4);
33          elseif rd<cp(5), x0=(a(5)+b(5))/2; y0=cp(5);
34          elseif rd<cp(6), x0=(a(6)+b(6))/2; y0=cp(6);
35          elseif rd<cp(7), x0=(a(7)+b(7))/2; y0=cp(7);
36          elseif rd<cp(8), x0=(a(8)+b(8))/2; y0=cp(8);
37          elseif rd<cp(9), x0=(a(9)+b(9))/2; y0=cp(9);
38          else
39              x0=b(10);
40              y0=cp(10);
41              x=x0+(rd-y0)/m(10);
42          end
43

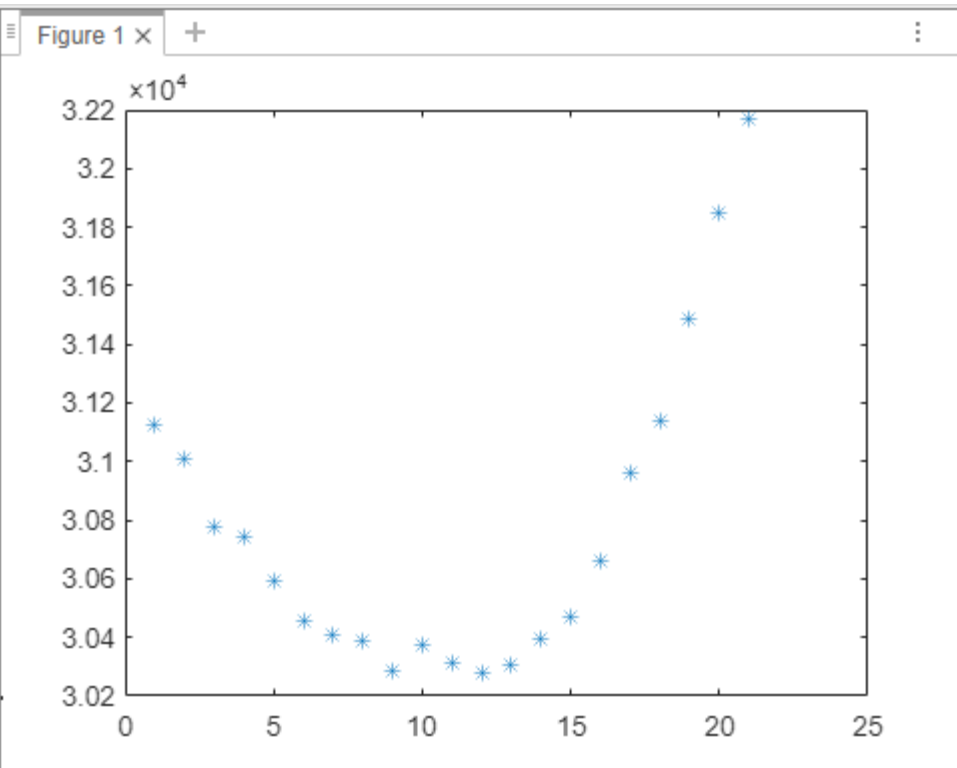
```

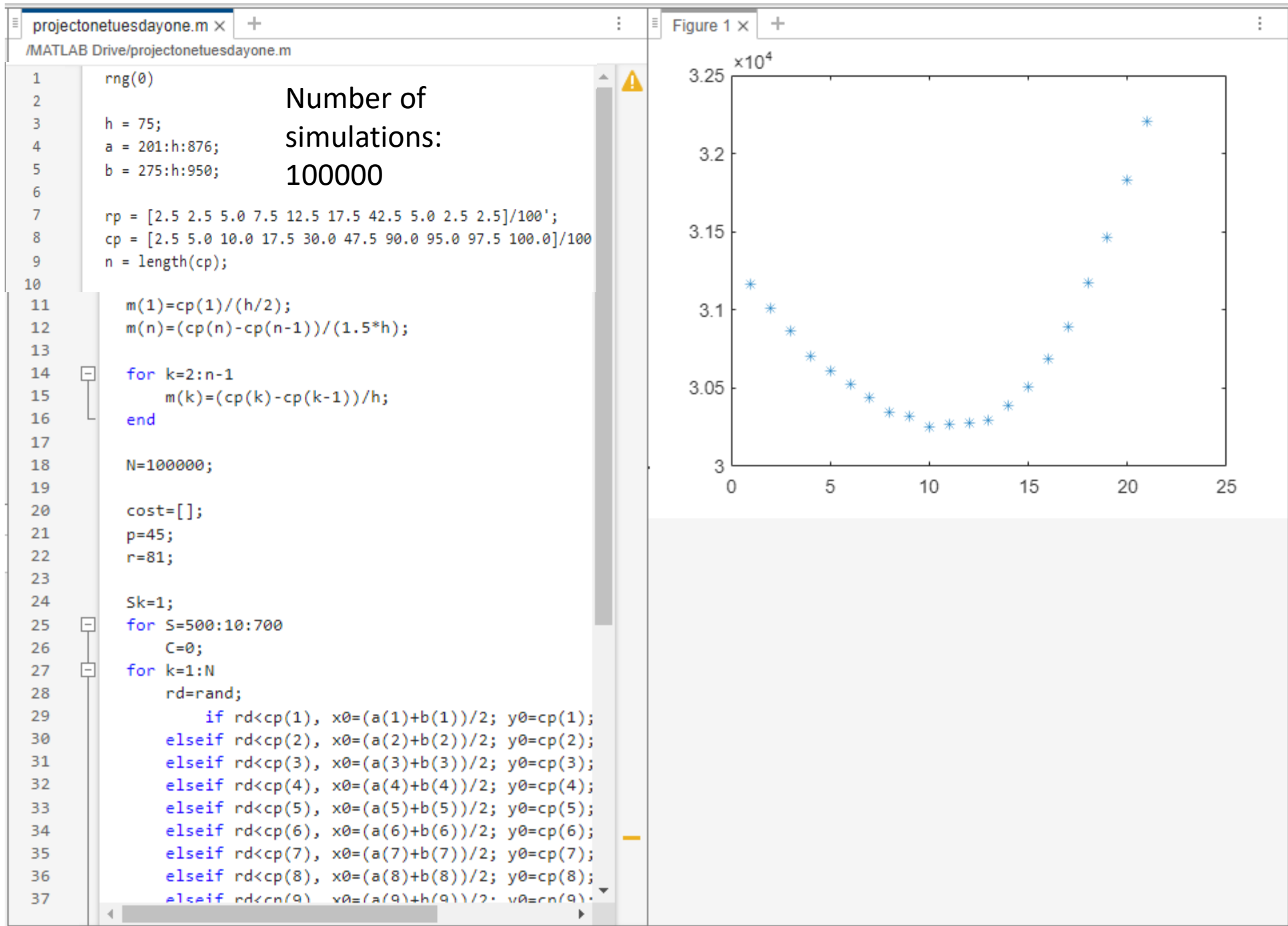
Number of
simulations: 3000

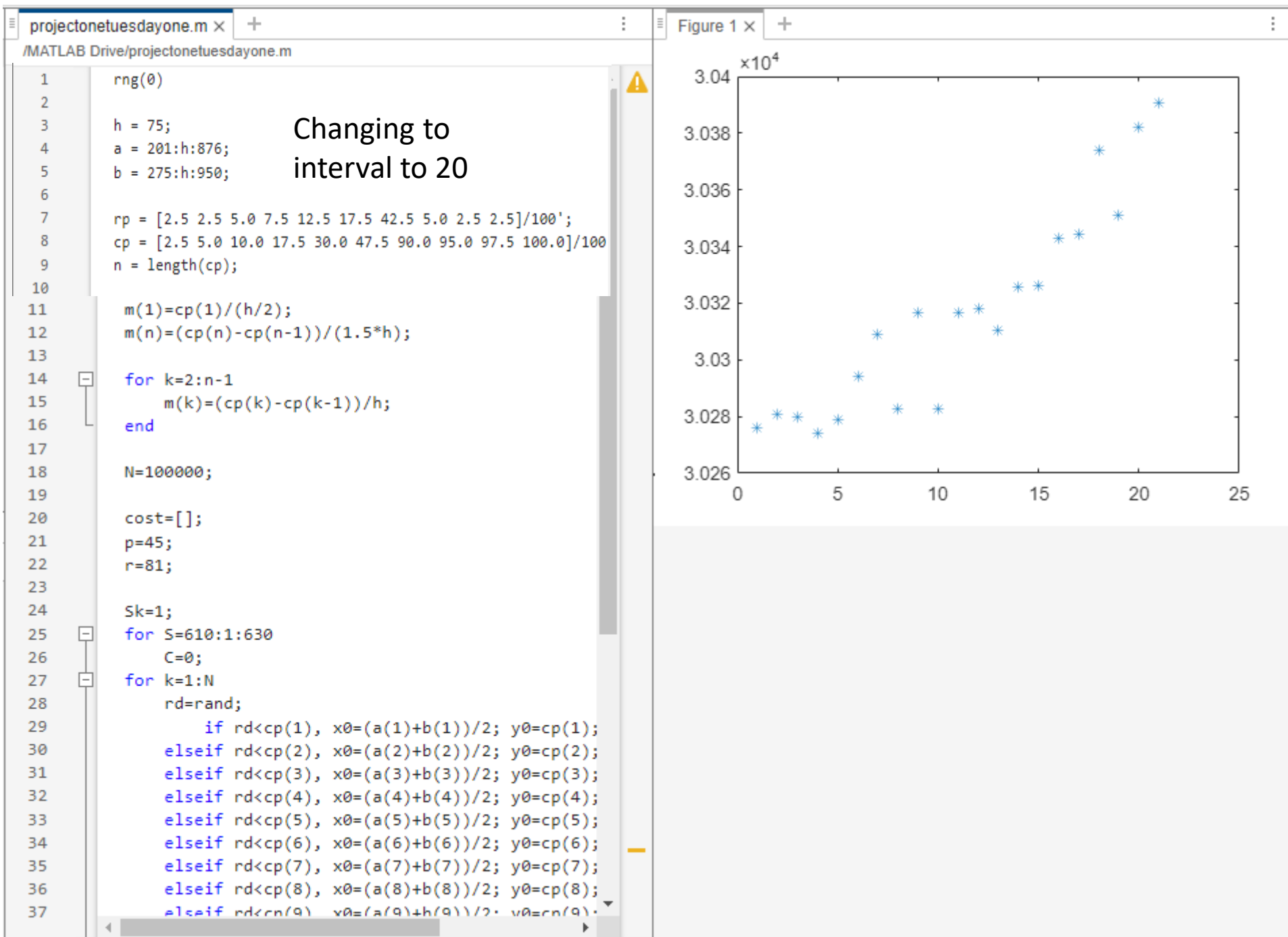


```
projectonetuesdayone.m x +
/MATLAB Drive/projectonetuesdayone.m
1  rng(0)
2
3  h = 75;
4  a = 201:h:876;
5  b = 275:h:950;
6
7  rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8  cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9  n = length(cp);
10
11  m(1)=cp(1)/(h/2);
12  m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14  for k=2:n-1
15      m(k)=(cp(k)-cp(k-1))/h;
16  end
17
18  N=10000;
19
20  cost=[];
21  p=45;
22  r=81;
23
24  Sk=1;
25  for S=500:10:700
26      C=0;
27      for k=1:N
28          rd=rand;
29          if rd<cp(1), x0=(a(1)+b(1))/2; y0=cp(1);
30          elseif rd<cp(2), x0=(a(2)+b(2))/2; y0=cp(2);
31          elseif rd<cp(3), x0=(a(3)+b(3))/2; y0=cp(3);
32          elseif rd<cp(4), x0=(a(4)+b(4))/2; y0=cp(4);
33          elseif rd<cp(5), x0=(a(5)+b(5))/2; y0=cp(5);
34          elseif rd<cp(6), x0=(a(6)+b(6))/2; y0=cp(6);
35          elseif rd<cp(7), x0=(a(7)+b(7))/2; y0=cp(7);
36          elseif rd<cp(8), x0=(a(8)+b(8))/2; y0=cp(8);
37          elseif rd<cp(9), x0=(a(9)+b(9))/2; y0=cp(9);
```

Number of
simulations: 10000







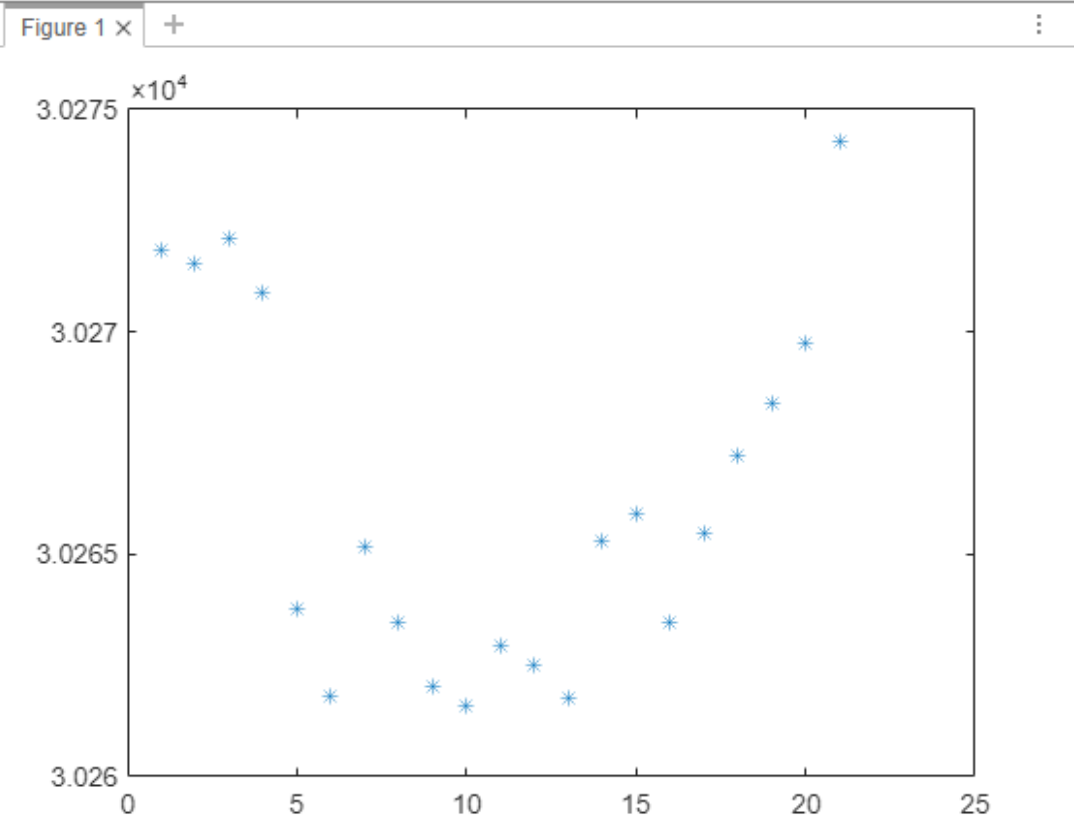
```

projectonetuesdayone.m * x
/MATLAB Drive/projectonetuesdayone.m

1  rng(0)
2
3  h = 75;
4  a = 201:h:876;
5  b = 275:h:950;
6
7  rp = [2.5 2.5 5.0 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8  cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9  n = length(cp);
10
11  m(1)=cp(1)/(h/2);
12  m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14  for k=2:n-1
15      m(k)=(cp(k)-cp(k-1))/h;
16  end
17
18  N=15000000;
19
20  cost=[];
21  p=45;
22  r=81;
23
24  Sk=1;
25  for S=590:1:610
26      C=0;
27      for k=1:N
28          rd=rand;
29          if rd<cp(1), x0=(a(1)+b(1))/2; y0=cp(1); x=x0+(rd-y0)
30          elseif rd<cp(2), x0=(a(2)+b(2))/2; y0=cp(2); x=x0+(rd-y0)
31          elseif rd<cp(3), x0=(a(3)+b(3))/2; y0=cp(3); x=x0+(rd-y0)
32          elseif rd<cp(4), x0=(a(4)+b(4))/2; y0=cp(4); x=x0+(rd-y0)
33          elseif rd<cp(5), x0=(a(5)+b(5))/2; y0=cp(5); x=x0+(rd-y0)
34          elseif rd<cp(6), x0=(a(6)+b(6))/2; y0=cp(6); x=x0+(rd-y0)
35          elseif rd<cp(7), x0=(a(7)+b(7))/2; y0=cp(7); x=x0+(rd-y0)
36          elseif rd<cp(8), x0=(a(8)+b(8))/2; y0=cp(8); x=x0+(rd-y0)
37          elseif rd<cp(9), x0=(a(9)+b(9))/2; y0=cp(9); x=x0+(rd-y0)
38          else
39              x0=b(10);
40              y0=cp(10);

```

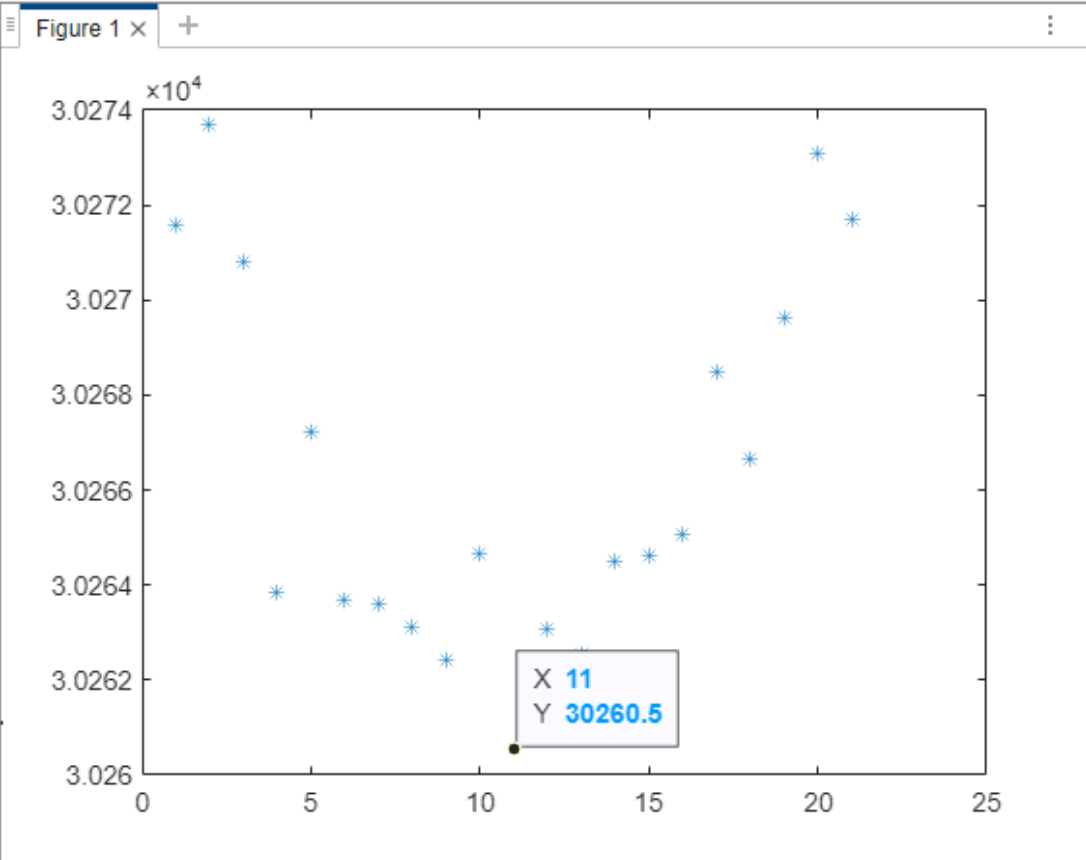
Number of
simulations:
15000000



```
projectonetuesdayone.m x +
/MATLAB Drive/projectonetuesdayone.m

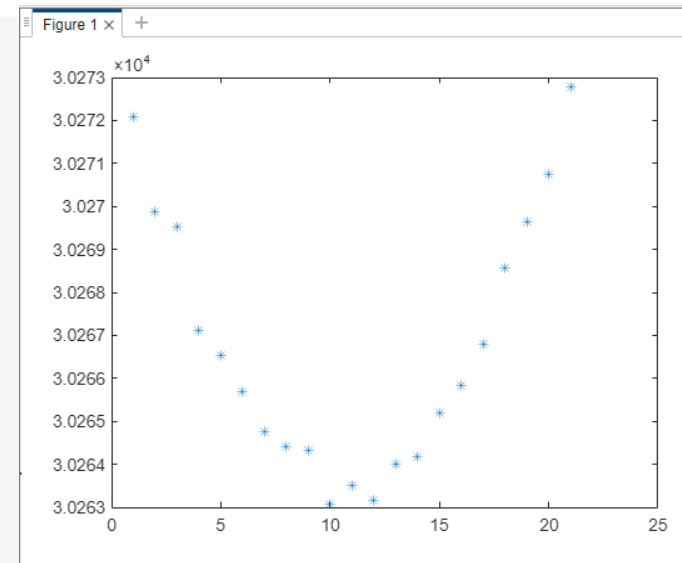
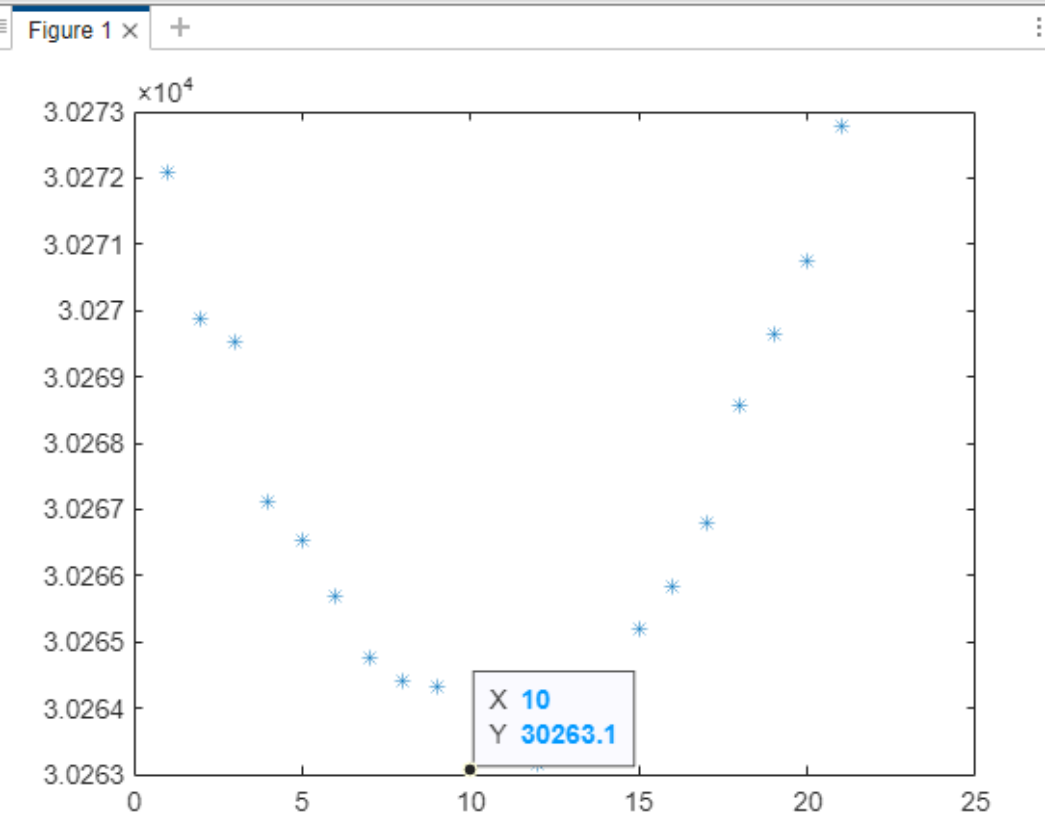
1   rng(0)
2
3   h = 75;
4   a = 201:h:876;
5   b = 275:h:950;
6
7   rp = [2.5 2.5 5.0 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8   cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9   n = length(cp);
10
11  m(1)=cp(1)/(h/2);
12  m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14  for k=2:n-1
15      m(k)=(cp(k)-cp(k-1))/h;
16  end
17
18  N=100000000;
19
20  cost=[];
21  p=45;
22  r=81;
23
24  Sk=1;
25  for S=590:1:610
26      C=0;
27      for k=1:N
28          rd=rand;
29          if rd<cp(1), x0=(a(1)+b(1))/2; y0=cp(1); x=x0+(rd-y0)
30          elseif rd<cp(2), x0=(a(2)+b(2))/2; y0=cp(2); x=x0+(rd-y0)
31          elseif rd<cp(3), x0=(a(3)+b(3))/2; y0=cp(3); x=x0+(rd-y0)
32          elseif rd<cp(4), x0=(a(4)+b(4))/2; y0=cp(4); x=x0+(rd-y0)
33          elseif rd<cp(5), x0=(a(5)+b(5))/2; y0=cp(5); x=x0+(rd-y0)
34          elseif rd<cp(6), x0=(a(6)+b(6))/2; y0=cp(6); x=x0+(rd-y0)
35          elseif rd<cp(7), x0=(a(7)+b(7))/2; y0=cp(7); x=x0+(rd-y0)
36          elseif rd<cp(8), x0=(a(8)+b(8))/2; y0=cp(8); x=x0+(rd-y0)
37          elseif rd<cp(9), x0=(a(9)+b(9))/2; y0=cp(9); x=x0+(rd-y0)
38          else
39              x0=b(10);
40              y0=cp(10);
```

Number of
simulations:
100000000



```
projectonetuesdayone.m x +
/MATLAB Drive/projectonetuesdayone.m

1   rng(0)
2
3   h = 75;
4   a = 201:h:876;
5   b = 275:h:950;
6
7   rp = [2.5 2.5 5.0 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8   cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9   n = length(cp);
10
11   m(1)=cp(1)/(h/2);
12   m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14   for k=2:n-1
15       m(k)=(cp(k)-cp(k-1))/h;
16   end
17
18   N=100000000;
19
20   cost=[];
21   p=45;
22   r=81;
23
24   Sk=1;
25   for S=590:1:610
26       C=0;
27       for k=1:N
28           rd=rand;
29           if rd<cp(1), x0=(a(1)+b(1))/2; y0=cp(1); x=x0+(rd-y0)
30           elseif rd<cp(2), x0=(a(2)+b(2))/2; y0=cp(2); x=x0+(rd-y0)
31           elseif rd<cp(3), x0=(a(3)+b(3))/2; y0=cp(3); x=x0+(rd-y0)
32           elseif rd<cp(4), x0=(a(4)+b(4))/2; y0=cp(4); x=x0+(rd-y0)
33           elseif rd<cp(5), x0=(a(5)+b(5))/2; y0=cp(5); x=x0+(rd-y0)
34           elseif rd<cp(6), x0=(a(6)+b(6))/2; y0=cp(6); x=x0+(rd-y0)
35           elseif rd<cp(7), x0=(a(7)+b(7))/2; y0=cp(7); x=x0+(rd-y0)
36           elseif rd<cp(8), x0=(a(8)+b(8))/2; y0=cp(8); x=x0+(rd-y0)
37           elseif rd<cp(9), x0=(a(9)+b(9))/2; y0=cp(9); x=x0+(rd-y0)
38           else
39               x0=b(10);
40               y0=cp(10);
```



What did the Part C results mean?

- The number from part C is expected to have a larger dollar amount since the substitutes get paid more, skewing the formula toward the more expensive side. However, there are fewer total teachers at 600 because there are fewer resources to hire more teachers if substitutes get paid more per teacher.
- Here, the least expected pay amount is estimated by the model to be \$30263.10 for 600 teachers.

```

projectonetuesdayone.m × projectonetuesdaytwo.m × +
/MATLAB Drive/projectonetuesdayone.m

1      rng(0)
2
3      h = 75;
4      a = 201:h:876;
5      b = 275:h:950;
6
7      rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8      cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100 ;
9      n = length(cp);
10
11     m(1)=cp(1)/(h/2);
12     m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14     for k=2:n-1
15         m(k)=(cp(k)-cp(k-1))/h;
16     end
17
18     N=1000;
19
20     cost=[];
21     p=45;
22     r=81;
23
24     Sk=1;
25     for S=100:100:900
26         C=0;
27         for k=1:N
28             rd=rand;
29             if rd<cp(1), x0=(a(1)+b(1))/2; y0=cp(1); x=x0+(rd-y0)/m(1);
30             elseif rd<cp(2), x0=(a(2)+b(2))/2; y0=cp(2); x=x0+(rd-y0)/m(2);
31             elseif rd<cp(3), x0=(a(3)+b(3))/2; y0=cp(3); x=x0+(rd-y0)/m(3);
32             elseif rd<cp(4), x0=(a(4)+b(4))/2; y0=cp(4); x=x0+(rd-y0)/m(4);
33             elseif rd<cp(5), x0=(a(5)+b(5))/2; y0=cp(5); x=x0+(rd-y0)/m(5);
34             elseif rd<cp(6), x0=(a(6)+b(6))/2; y0=cp(6); x=x0+(rd-y0)/m(6);
35             elseif rd<cp(7), x0=(a(7)+b(7))/2; y0=cp(7); x=x0+(rd-y0)/m(7);
36             elseif rd<cp(8), x0=(a(8)+b(8))/2; y0=cp(8); x=x0+(rd-y0)/m(8);
37             elseif rd<cp(9), x0=(a(9)+b(9))/2; y0=cp(9); x=x0+(rd-y0)/m(9);

```

Base script code for 5.54PJ. Part C

```

else
    x0=b(10);
    y0=cp(10);
    x=x0+(rd-y0)/m(10);
end

if x<S, Cxs=p*S;
else
    Cxs=p*S + (x-S)*r;
end
C=C + Cxs;

end
Cost(Sk) = C/N;
Sk=Sk + 1;
end
plot(Cost, '*')

```

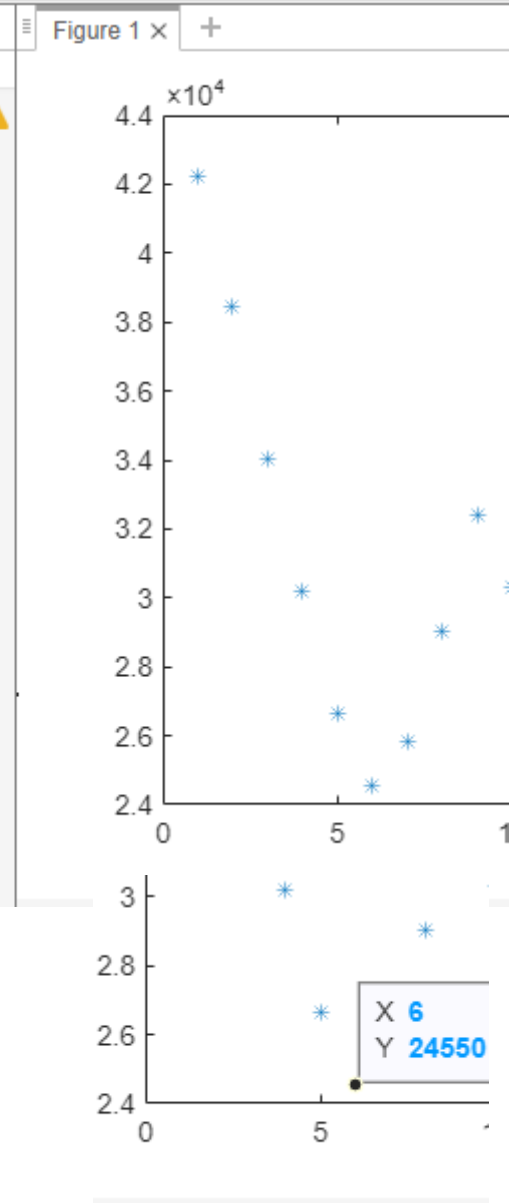
Here, we will find the optimum values for teacher pay rates and number pools in Los Angeles School District.

- In the first scenario, pay rates will be set to substitute teachers earning \$36 and full-time teachers earning \$81.
- The range of the number of teachers from (0,1000).
- The focus will be on Tuesdays.
- We'll find the least expected costs.
- Assumptions: All teachers work the same hours throughout, and there's no overtime. You can't hire all substitute teachers to replace full-time teachers.
- We'll increase the number of simulations to fine-tune the model.

```
projectonetuesdaytwo.m × +
/MATLAB Drive/projectonetuesdaytwo.m

1  rng(0)
2
3  h = 75;
4  a = 201:h:876;
5  b = 275:h:950;
6
7  rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8  cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9  n = length(cp);
10
11  m(1)=cp(1)/(h/2);
12  m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14  for k=2:n-1
15      m(k)=(cp(k)-cp(k-1))/h;
16  end
17
18  N=1000;
19
20  cost=[];
21  p=36;
22  r=81;
23
24  Sk=1;
```

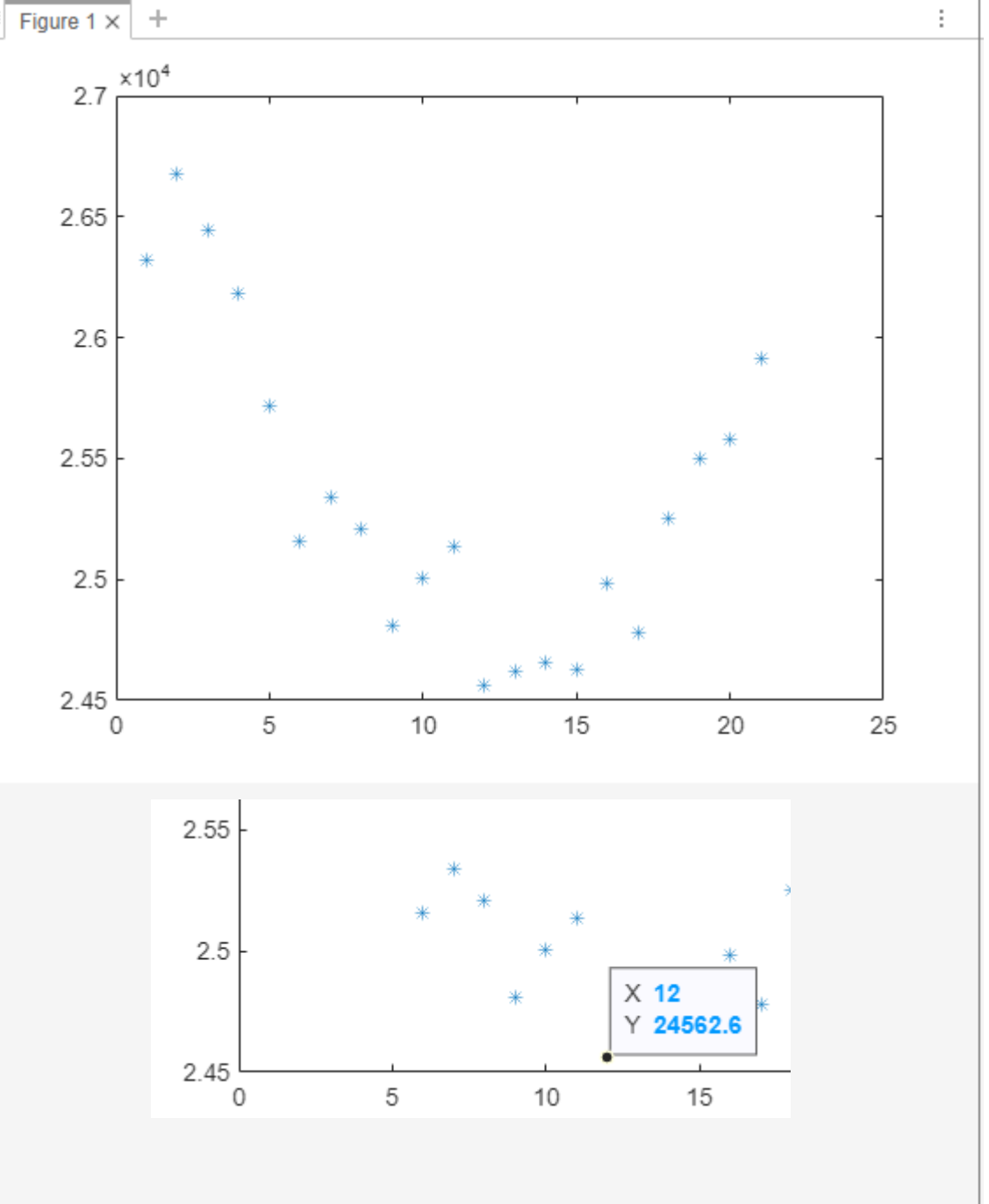
Number of
simulations: 1000



```
projectonetuesdaytwo.m x +
/MATLAB Drive/projectonetuesdaytwo.m

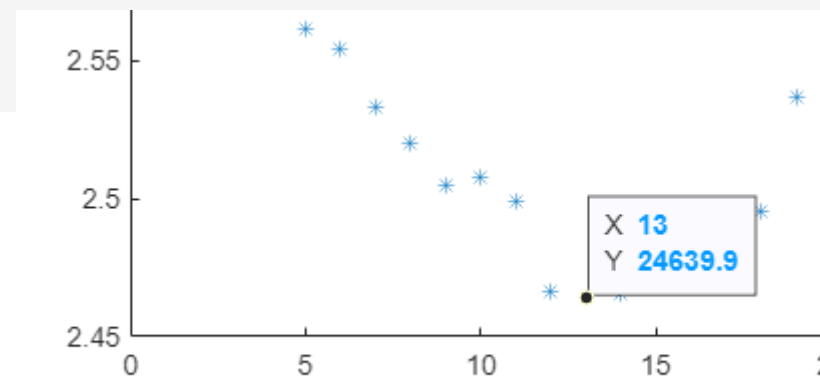
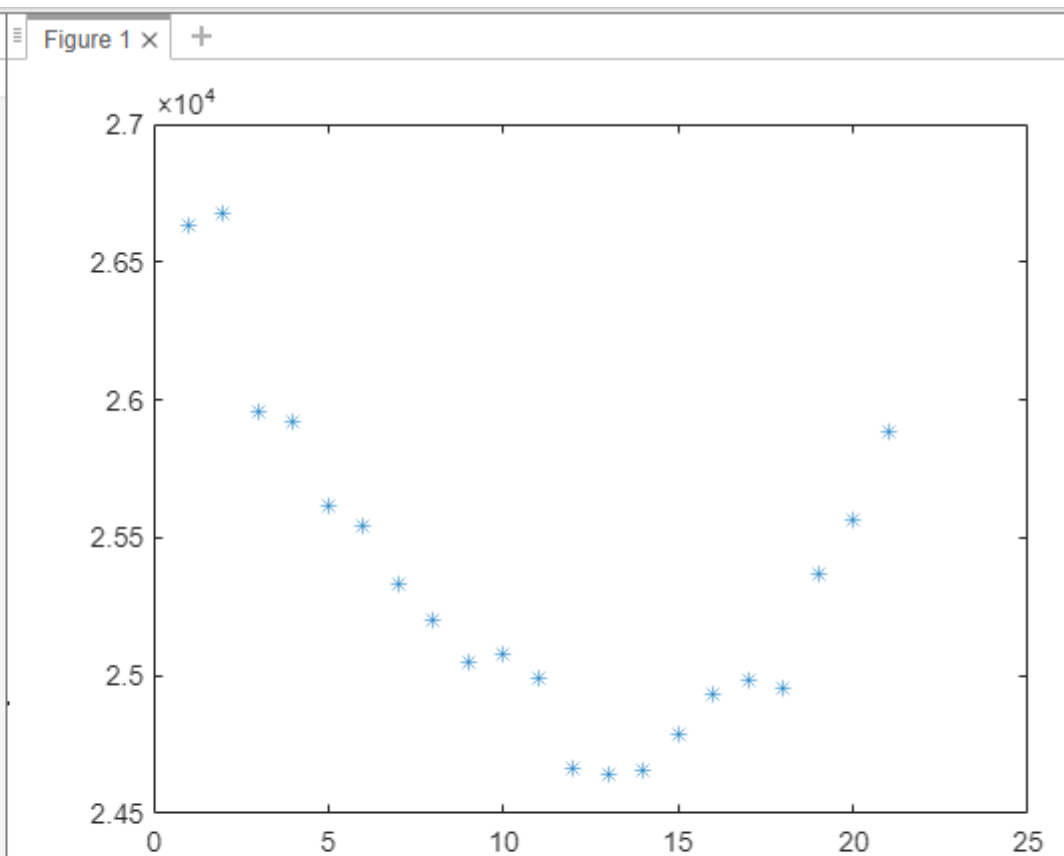
1  rng(0)
2
3  h = 75;
4  a = 201:h:876;
5  b = 275:h:950;
6
7  rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8  cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9  n = length(cp);
10
11  m(1)=cp(1)/(h/2);
12  m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14  for k=2:n-1
15      m(k)=(cp(k)-cp(k-1))/h;
16  end
17
18  N=1000;
19
20  cost=[];
21  p=36;
22  r=81;
23
24  Sk=1;
25  for S=500:10:700
26      C=0;
27      for k=1:N
28          rd=rand;
29          if rd<cp(1), x0=(a(1)+b(1))/2; y0=cp(1); x=x0+(rd-y0)
30          elseif rd<cp(2), x0=(a(2)+b(2))/2; y0=cp(2); x=x0+(rd-y0)
31          elseif rd<cp(3), x0=(a(3)+b(3))/2; y0=cp(3); x=x0+(rd-y0)
32          elseif rd<cp(4), x0=(a(4)+b(4))/2; y0=cp(4); x=x0+(rd-y0)
33          elseif rd<cp(5), x0=(a(5)+b(5))/2; y0=cp(5); x=x0+(rd-y0)
34          elseif rd<cp(6), x0=(a(6)+b(6))/2; y0=cp(6); x=x0+(rd-y0)
35          elseif rd<cp(7), x0=(a(7)+b(7))/2; y0=cp(7); x=x0+(rd-y0)
36          elseif rd<cp(8), x0=(a(8)+b(8))/2; y0=cp(8); x=x0+(rd-y0)
37          elseif rd<cp(9), x0=(a(9)+b(9))/2; y0=cp(9); x=x0+(rd-y0)
```

Number of
simulations: 1000



```
projectonetuesdaytwo.m × +
/MATLAB Drive/projectonetuesdaytwo.m

1  rng(0)
2
3  h = 75;
4  a = 201:h:876;
5  b = 275:h:950;
6
7  rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8  cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9  n = length(cp);
10
11  m(1)=cp(1)/(h/2);
12  m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14  for k=2:n-1
15      m(k)=(cp(k)-cp(k-1))/h;
16  end
17
18  N=2000;
19
20  cost=[];
21  p=36;
22  r=81;
23
24  Sk=1;
25  for S=500:10:700
26      C=0;
27      for k=1:N
28          rd=rand;
```

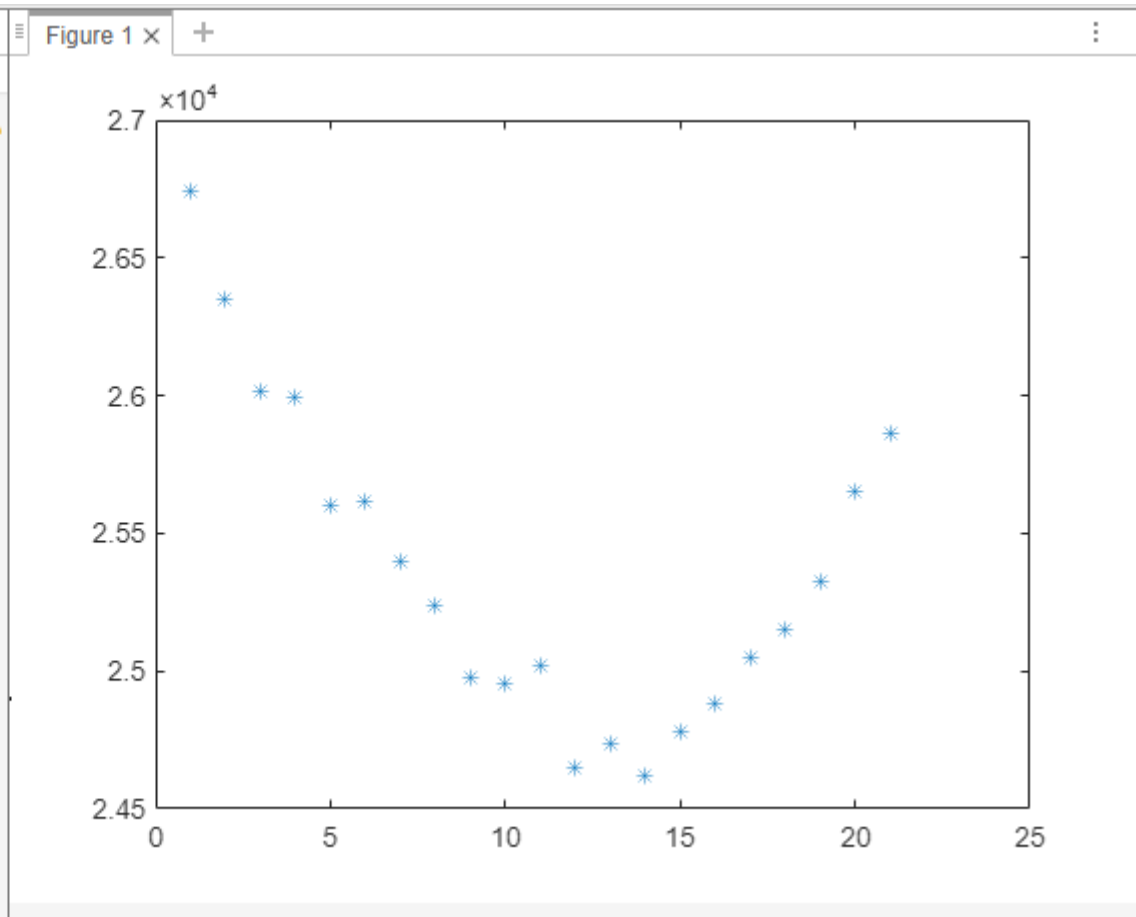


Number of
simulations: 2000

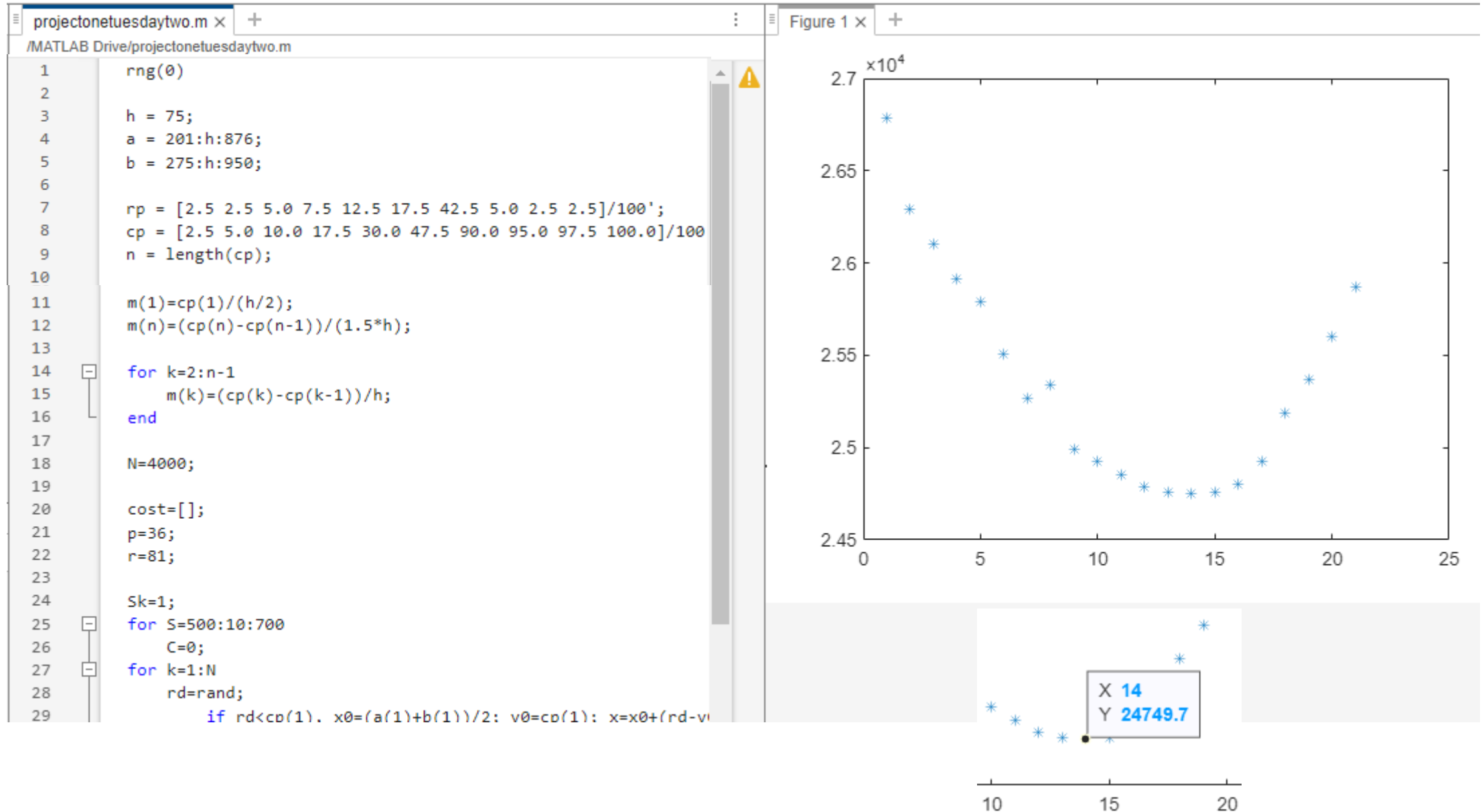
```
projectonetuesdaytwo.m x +
/MATLAB Drive/projectonetuesdaytwo.m

1  rng(0)
2
3  h = 75;
4  a = 201:h:876;
5  b = 275:h:950;
6
7  rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8  cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9  n = length(cp);
10
11  m(1)=cp(1)/(h/2);
12  m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14  for k=2:n-1
15      m(k)=(cp(k)-cp(k-1))/h;
16  end
17
18  N=3000;
19
20  cost=[];
21  p=36;
22  r=81;
23
24  Sk=1;
```

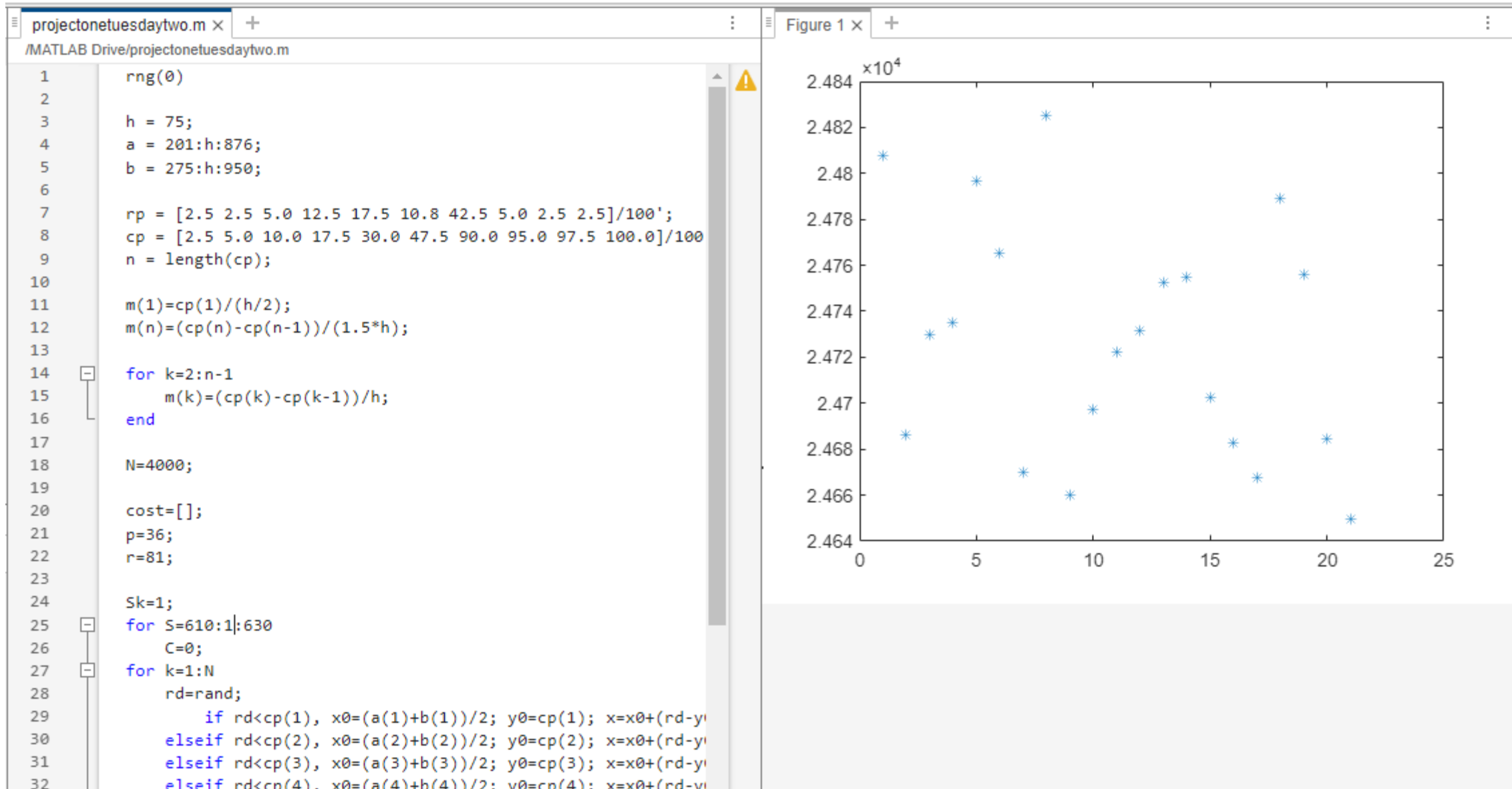
Number of
simulations: 3000



Number of
simulations: 4000



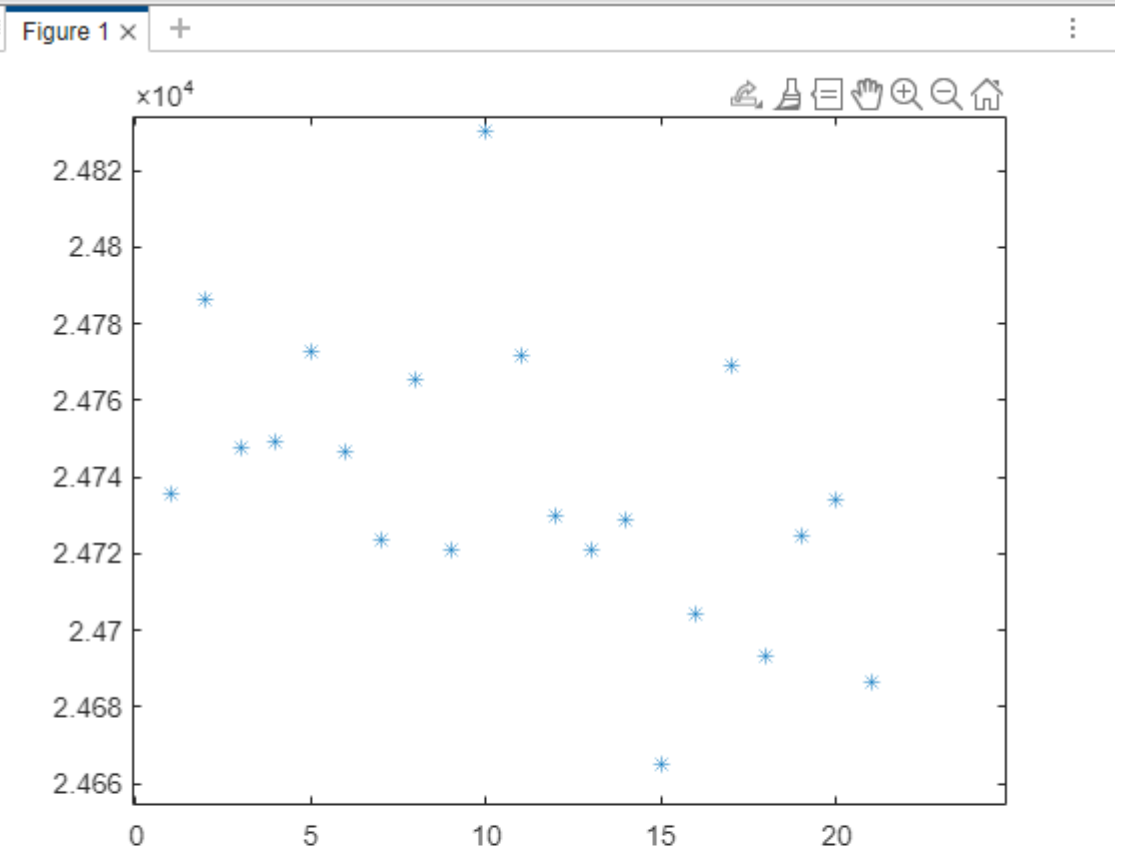
Looking at interval:20



Number of
simulations: 10000

```
projectonetuesdaytwo.m x +
/MATLAB Drive/projectonetuesdaytwo.m

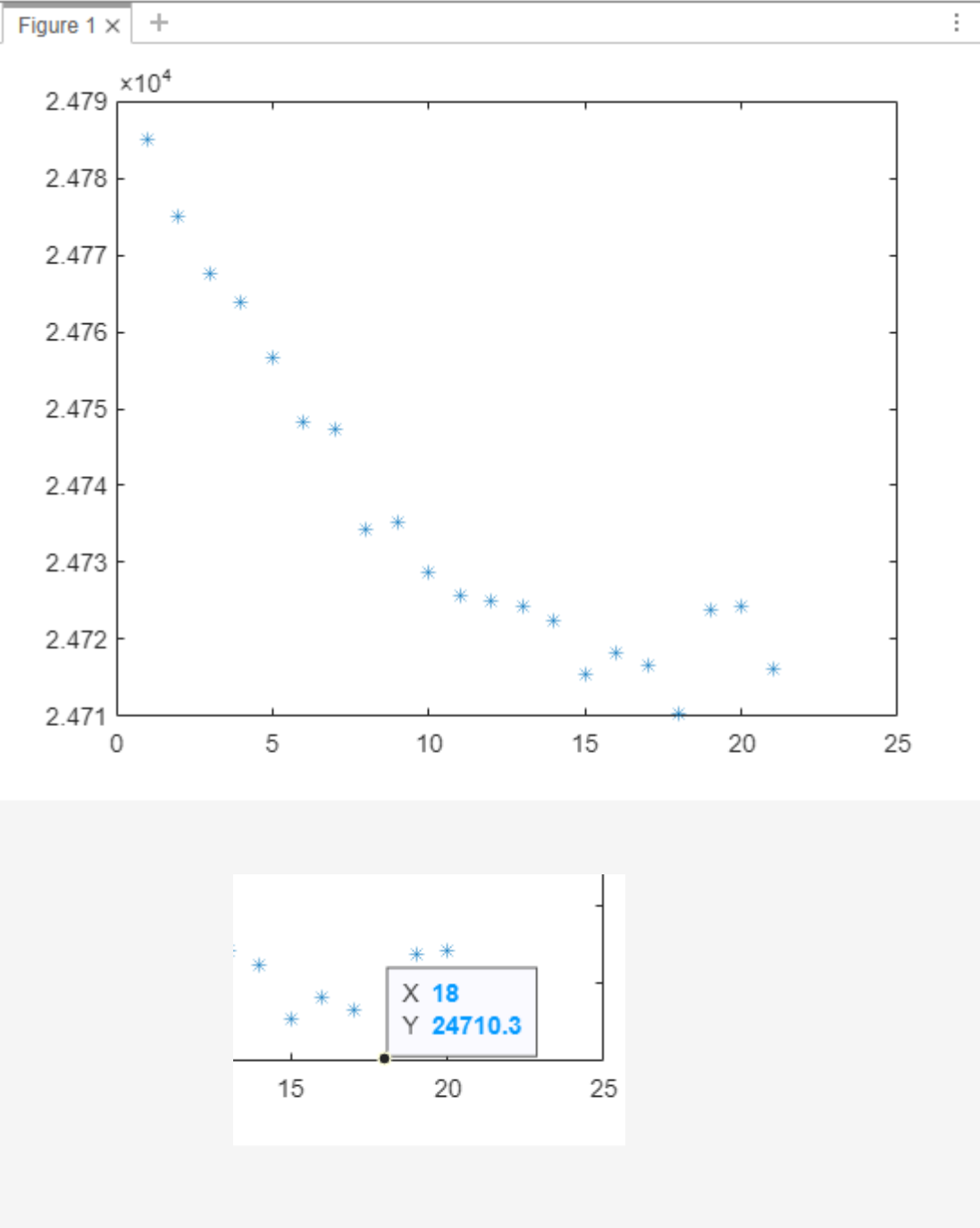
1  rng(0)
2
3  h = 75;
4  a = 201:h:876;
5  b = 275:h:950;
6
7  rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8  cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9  n = length(cp);
10
11  m(1)=cp(1)/(h/2);
12  m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14  for k=2:n-1
15      m(k)=(cp(k)-cp(k-1))/h;
16  end
17
18  N=10000;
19
20  cost=[];
21  p=36;
22  r=81;
23
24  Sk=1;
25  for S=610:1:630
26      C=0;
27      for k=1:N
28          rd=rand;
29          if rd<cp(1), x0=(a(1)+b(1))/2; y0=cp(1); x=x0+(rd-y0)
30          elseif rd<cp(2), x0=(a(2)+b(2))/2; y0=cp(2); x=x0+(rd-y0)
31          elseif rd<cp(3), x0=(a(3)+b(3))/2; y0=cp(3); x=x0+(rd-y0)
```



```
projectonetuesdaytwo.m x +
/MATLAB Drive/projectonetuesdaytwo.m

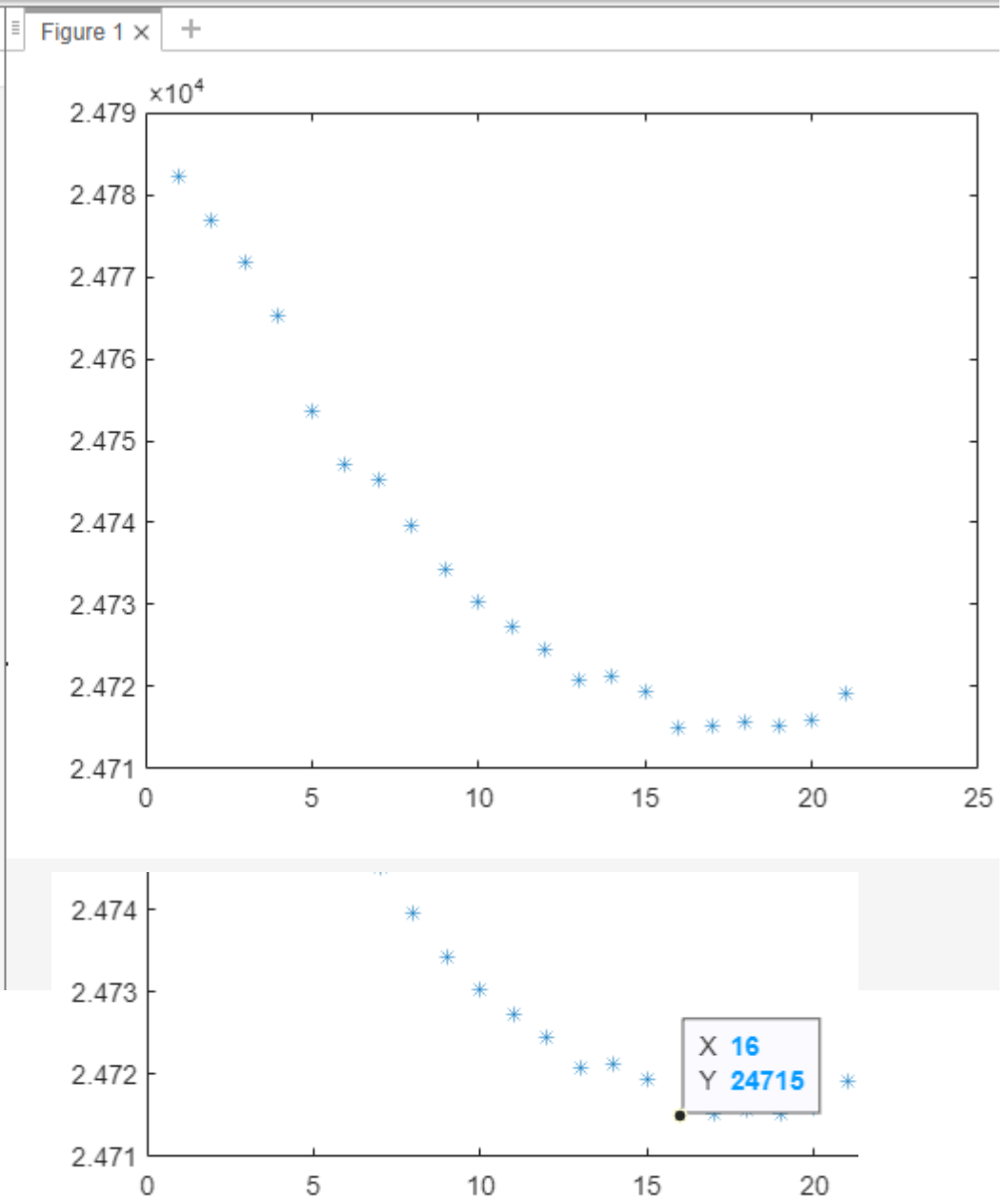
1 rng(0)
2
3 h = 75;
4 a = 201:h:876;
5 b = 275:h:950;
6
7 rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8 cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9 n = length(cp);
10
11 m(1)=cp(1)/(h/2);
12 m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14 for k=2:n-1
15     m(k)=(cp(k)-cp(k-1))/h;
16 end
17
18 N=1000000;
19
20 cost=[];
21 p=36;
22 r=81;
23
24 Sk=1;
25 for S=610:1:630
26     C=0;
27     for k=1:N
28         rd=rand;
29         if rd<cp(1), x0=(a(1)+b(1))/2; y0=cp(1); x=x0+(rd-y0)
30         elseif rd<cp(2), x0=(a(2)+b(2))/2; y0=cp(2); x=x0+(rd-y0)
31         elseif rd<cp(3), x0=(a(3)+b(3))/2; y0=cp(3); x=x0+(rd-y0)
32         elseif rd<cp(4), x0=(a(4)+b(4))/2; y0=cp(4); x=x0+(rd-y0)
33         elseif rd<cp(5), x0=(a(5)+b(5))/2; y0=cp(5); x=x0+(rd-y0)
34         elseif rd<cp(6), x0=(a(6)+b(6))/2; y0=cp(6); x=x0+(rd-y0)
35         elseif rd<cp(7), x0=(a(7)+b(7))/2; y0=cp(7); x=x0+(rd-y0)
36         elseif rd<cp(8), x0=(a(8)+b(8))/2; y0=cp(8); x=x0+(rd-y0)
37         elseif rd<cp(9), x0=(a(9)+b(9))/2; y0=cp(9); x=x0+(rd-y0)
```

Number of
simulations:
1000000



```
projectonetuesdaytwo.m x +
/MATLAB Drive/projectonetuesdaytwo.m
1 rng(0)
2
3 h = 75;
4 a = 201:h:876;
5 b = 275:h:950;
6
7 rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8 cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100
9 n = length(cp);
10
11 end
12
13 N=10000000;
14
15 cost=[];
16 p=36;
17 r=81;
18
19 Sk=1;
20 for S=610:1:630
21     C=0;
22     for k=1:N
23         rd=rand;
```

Number of
simulations:
10000000



```

1  rng(0)
2
3  h = 75;
4  a = 201:h:876;
5  b = 275:h:950;
6
7  rp = [2.5 2.5 5.0 7.5 12.5 17.5 42.5 5.0 2.5 2.5]/100';
8  cp = [2.5 5.0 10.0 17.5 30.0 47.5 90.0 95.0 97.5 100.0]/100';
9  n = length(cp);
10
11  m(1)=cp(1)/(h/2);
12  m(n)=(cp(n)-cp(n-1))/(1.5*h);
13
14  for k=2:n-1
15      m(k)=(cp(k)-cp(k-1))/h;
16  end
17
18  N=10000000;
19
20  cost=[];
21  p=36;
22  r=81;
23
24  Sk=1;
25  for S=610:1:630
26      C=0;
27      for k=1:N
28          rd=rand;
29          if rd<cp(1), x0=(a(1)+b(1))/2; y0=cp(1); x=x0+(rd-y0)/m(1);
30          elseif rd<cp(2), x0=(a(2)+b(2))/2; y0=cp(2); x=x0+(rd-y0)/m(2);
31          elseif rd<cp(3), x0=(a(3)+b(3))/2; y0=cp(3); x=x0+(rd-y0)/m(3);
32          elseif rd<cp(4), x0=(a(4)+b(4))/2; y0=cp(4); x=x0+(rd-y0)/m(4);
33          elseif rd<cp(5), x0=(a(5)+b(5))/2; y0=cp(5); x=x0+(rd-y0)/m(5);
34          elseif rd<cp(6), x0=(a(6)+b(6))/2; y0=cp(6); x=x0+(rd-y0)/m(6);
35          elseif rd<cp(7), x0=(a(7)+b(7))/2; y0=cp(7); x=x0+(rd-y0)/m(7);
36          elseif rd<cp(8), x0=(a(8)+b(8))/2; y0=cp(8); x=x0+(rd-y0)/m(8);
37          elseif rd<cp(9), x0=(a(9)+b(9))/2; y0=cp(9); x=x0+(rd-y0)/m(9);
38          else
39              x0=b(10);
40              y0=cp(10);

```

Part D script code

```

elseif rd<cp(9), x0=(a(9)+b(9))/2; y0=cp(9); x=x0+(rd-y0)/m(9);
else
    x0=b(10);
    y0=cp(10);
    x=x0+(rd-y0)/m(10);
end

if x<S, Cxs=p*S;
else
    Cxs=p*S + (x-S)*r;
end
C=C + Cxs;
end
Cost(Sk) = C/N;
Sk=Sk + 1;
end
plot(Cost, '*')

```

What did the Part D results mean?

- Here, the least expected pay amount is estimated by the model at \$24715.00 for 626 teachers.
- In part D, there are more teachers for less overall pay because substitutes are paid less, skewing the formula to cheap overall pay @626 teachers.

Part 5. Overall Simulation Conclusions

- The number from part C is expected to have a larger dollar amount since the substitutes get paid more, skewing the formula toward the more expensive side. However, there are fewer total teachers at 600 because there are fewer resources to hire more teachers if substitutes get paid more per teacher.
- In part D, there are more teachers for less overall pay because substitutes are paid less, skewing the formula to cheap overall pay @626 teachers.
- Intuitively, the model makes sense. If the pay for substitutes remains low, then the pay formula will get progressively cheaper to hire more teachers. Conversely, we have the same resources, and the total number of teachers will fall with higher-paid substitutes.